

HIERARCHICAL GENERALIZED LINEAR MODELS WITH R. APPLICATIONS IN PLANT/ANIMAL BREEDING AND GENETICS

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Recap

- The fixed and random effects in a linear mixed model can be estimated using weighted least squares on an augmented model.
- Hat values and other diagnostics developed for linear regression can be used.
- The variance components can be computed using Gamma GLM.
- The algorithm for fitting a linear mixed model iterates between
 - estimating fixed and random effects
 - estimating variance components

Course outline

- **Linear regression**
 - diagnostic plots
- **Weighted least squares**
- **Generalized linear models**
- **Linear mixed models**
 - Estimating fixed and random effects
 - It's a weighted least squares problem!
 - Estimating variance components
 - Application on repeated observations
 - Fitting an animal model using the additive relationship matrix
 - Smoothing

Pedigree data

A full-sib family example.

Animal	Father	Mother
1	.	.
2	.	.
3	1	2
4	1	2

Correlation matrix: “additive relationship matrix”

$$\mathbf{A} = \begin{pmatrix} 1 & 0 & 0.5 & 0.5 \\ 0 & 1 & 0.5 & 0.5 \\ 0.5 & 0.5 & 1 & 0.5 \\ 0.5 & 0.5 & 0.5 & 1 \end{pmatrix}$$

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Additive genetic effects

We have the model $y = \mathbf{X}\beta + \mathbf{Z}a + e$ where $V(a) = \mathbf{A}\sigma_a^2$ and $\mathbf{Z}$ is a 4×4 diagonal matrix with elements equal to 1.

The mixed model equations are

$$\begin{pmatrix} \mathbf{X}'\mathbf{X} & \mathbf{X}'\mathbf{Z} \\ \mathbf{Z}'\mathbf{X} & \mathbf{Z}'\mathbf{Z} + \frac{\sigma_e^2}{\sigma_a^2}\mathbf{A}^{-1} \end{pmatrix} \begin{pmatrix} \beta \\ a \end{pmatrix} = \begin{pmatrix} \mathbf{X}'y \\ \mathbf{Z}'y \end{pmatrix}$$

The random effects a are often referred to as *additive genetic effects* or *breeding values*.

The (narrow sense) heritability is computed as:

$$h^2 = \frac{\sigma_a^2}{\sigma_a^2 + \sigma_e^2}$$

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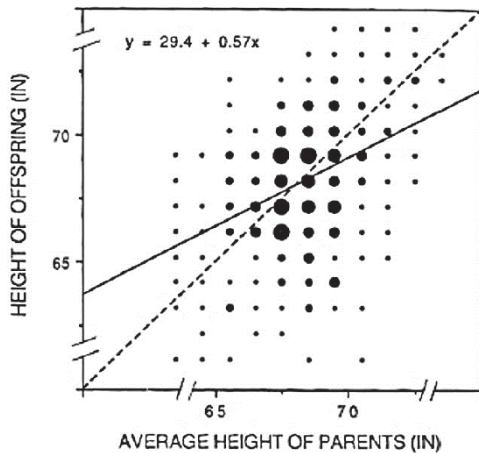
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Heritability



Missing trait values

Suppose individual 4 has no observed response value. Can we still estimate an additive genetic effect a_4 ?

Yes!!! Let

$$\mathbf{Z} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

then the vector a (of length 4) can still be computed from the mixed model equations

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where $\mathbf{A}$ is the same 4×4 matrix as before, and y has length 3.

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Equivalent animal models

The following two models are equivalent

$$y = \mathbf{X}\beta + \mathbf{1}a + e$$

$$y = \mathbf{X}\beta + \mathbf{Z}u + e$$

if $V(a) = \mathbf{A}\sigma_a^2$, $V(u) = \mathbf{I}\sigma_a^2$ and $\mathbf{Z}\mathbf{Z}^T = \mathbf{A}$.

Because

$$V(y) = V(\mathbf{1}a + e) = \mathbf{A}\sigma_a^2 + \mathbf{I}\sigma_e^2$$

and

$$V(y) = V(\mathbf{Z}u + e) = \mathbf{Z}\mathbf{Z}^T\sigma_a^2 + \mathbf{I}\sigma_e^2 = \mathbf{A}\sigma_a^2 + \mathbf{I}\sigma_e^2$$

Cholesky factorization

Let **L** be the **Cholesky factorization** of **A**. Then **L** is a lower triangular matrix having the property $\mathbf{LL}^T = \mathbf{A}$.

Suppose we have

$$\mathbf{A} = \begin{pmatrix} 1 & 0 & 0.5 & 0.5 \\ 0 & 1 & 0.5 & 0.5 \\ 0.5 & 0.5 & 1 & 0.5 \\ 0.5 & 0.5 & 0.5 & 1 \end{pmatrix}$$

then

$$\mathbf{L} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0.5 & 0.5 & \sqrt{0.5} & 0 \\ 0.5 & 0.5 & 0 & \sqrt{0.5} \end{pmatrix}$$

R code `L = t( chol(A) )`

NB: Works only for positive definite matrices (i.e. all eigenvalues > 0).

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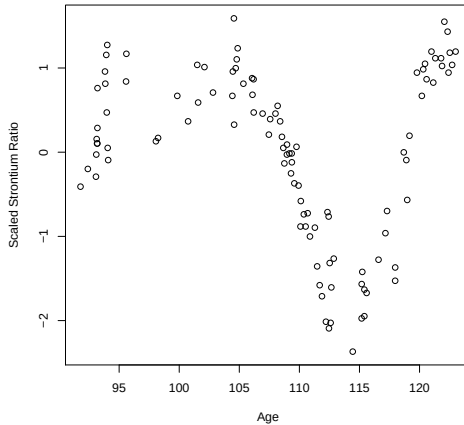
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Applying it in hglm

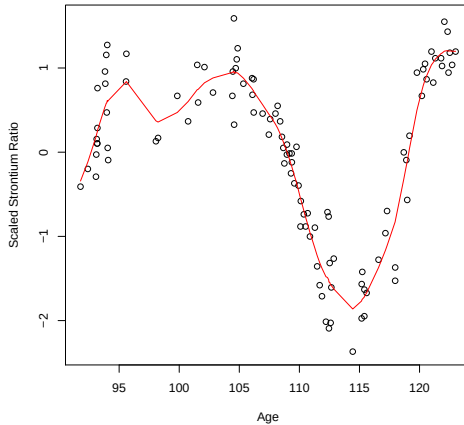
Let **L** be the **Cholesky factorization** of **A**. Then an animal model can be fitted in R using the hglm function as

```
hglm(y=y, X=X, Z=L)
```


Fossil Data Example



Fossil Data Example - fitted semi-parametric function



Uses a spatial covariance matrix

The Matern spatial model gives a correlation matrix.

$$\mathbf{C} = \begin{pmatrix} 1 & f(d_{12}) & f(d_{13}) & f(d_{14}) \\ f(d_{12}) & 1 & f(d_{23}) & f(d_{24}) \\ f(d_{13}) & f(d_{23}) & 1 & f(d_{34}) \\ f(d_{14}) & f(d_{24}) & f(d_{34}) & 1 \end{pmatrix}$$

where d_{ij} is the distance between observation i and j , and the correlations are $f(d_{ij}) = \exp(-d_{ij}/\rho)$ with a range parameter ρ .

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Applying it in hglm

Let **L** be the **Cholesky factorization** of **C**. Then the model can be fitted in R using the hglm function as

```
hglm(y=y, X=X, Z=L)
```

Effective number of parameters

Lee and Nelder (1996) used the concept of effective number of parameters in their original paper on HGLMs.

$$p_{eff} = \sum_{i=1}^n h_{ii}$$

Later Spiegelhalter et al. (2002) put it in a Bayesian setting.

R packages developed for semi-parametric smoothing

There are several packages in R developed for semi-parametric smoothing that can be put into two groups: *Generalized Additive Models* and *Gaussian Processes*.

- Generalized Additive Models
 - **gam**
 - **mgcv**
- Gaussian Processes
 - **GPfit**
 - and many others